

SESSION 3

AI in the Classroom

A Deep Dive into Google NotebookLM

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NotebookLM is an AI-powered research and writing assistant from Google.

It works exclusively from sources you provide — not the open internet.



Source-Grounded

Every answer is anchored to your uploaded materials — PDFs, docs, slides, links, or YouTube videos.



Cite-Verified

Responses include inline citations so you can trace any claim back to the exact passage in your sources.



Multi-Format Output

Transform your sources into podcasts, slide decks, reports, flashcards, quizzes, infographics, and more.



Three distinct workspaces — each with a focused purpose.

01



SOURCES

Your Knowledge Base

Upload and manage the documents, PDFs, slides, links, and YouTube videos that become the AI's exclusive reference library for your project. NotebookLM will only draw from what you upload — nothing else.

Library & Upload Hub

02



CHAT

Interactive Workspace

Query the AI in a conversational interface. Ask questions, request summaries, and explore your sources through dialogue. Every response includes inline citations — click any citation to jump to the exact passage.

Cited Q&A Interface

03



STUDIO

Creation Hub

Transform your source material into structured learning artifacts. Audio overviews, slide decks, video overviews, tailored reports, flashcards, quizzes, and infographics — all generated from your sources.

Output Generator

STUDIO OUTPUT CATEGORIES

What the Studio can generate from your sources



Audio Overview

AI podcast-style conversation about your sources. Interactive mode lets you steer the hosts' focus and tone in real time.



Slide Deck

Structured visual summary of your sources — formatted for classroom presentations or self-guided reading.



Video Overview

A visual walkthrough of key concepts and insights drawn from your uploaded materials.



Tailored Reports

Comprehensive documents customized for purpose — executive summaries, detailed analyses, briefing docs, and more.



Flashcards

Study cards for key terms and concepts — great for exam prep or vocabulary review.



Quiz

Auto-generated questions based on your materials to test comprehension and retention.



Infographic

Visual data representations including charts, timelines, or concept maps — generated via steering prompts.

INSTRUCTOR NOTE: HIDDEN FEATURES



Mind Maps and Data Tables are not standalone Studio buttons.

They can be generated inside the Infographic or Tailored Report features by providing specific steering prompts — just tell the AI what structure or format you want.

Mind Map

Infographic feature

Steering Prompt Example:

Try: "Create a mind map of the main themes in my sources, with branches for subtopics."

Data Table

Tailored Report feature

Steering Prompt Example:

Try: "Generate a comparison table of the key concepts across all my sources."

SPOTLIGHT: AUDIO OVERVIEW

NotebookLM's most distinctive feature — and its most misunderstood.



What It Is

Two AI hosts have a genuine podcast-style conversation about your uploaded sources.

Tone: Conversational, engaging, and naturally paced.

Grounding: All discussion is drawn exclusively from your sources — not the web.

Length: Typically 10–20 minutes depending on source depth.

Interactive Mode

While the podcast plays, you can interrupt the hosts and give them live instructions:

"Focus more on the methodology section."

"Explain that concept in simpler terms."

"Ask each other what you found most surprising."

"Wrap up and summarize the main takeaways."

CLASSROOM APPLICATIONS

How faculty and students can put NotebookLM to work



Faculty

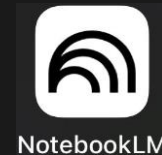
- Upload course readings and generate a podcast overview for pre-class listening
- Create auto-graded quizzes from your own course materials
- Generate a slide deck summary from a dense academic article
- Build a tailored executive briefing from multiple research papers
- Create student study guides from lecture notes and readings



Students

- Upload all readings and use Chat to ask clarifying questions with citations
- Generate flashcards for exam review from assigned texts
- Listen to an Audio Overview while commuting or exercising
- Create a structured report from research sources for a paper outline
- Use the quiz feature to self-test before exams

KEY TAKEAWAYS



What to remember about NotebookLM

1

NotebookLM is source-grounded. It only knows what you give it — and that's a feature, not a bug. It keeps AI responses accurate and attributable.

2

The three sections have distinct purposes: Sources (upload), Chat (query with citations), Studio (generate outputs). Know which workspace to be in.

3

The Audio Overview is interactive. You can steer the hosts mid-conversation — this is one of NotebookLM's most powerful and underused capabilities.

4

Mind Maps and Data Tables are hidden features. They live inside Infographic and Tailored Report — unlock them with steering prompts.